

EC200U&EG91xU Series QuecOpen Data Call API Reference Manual

LTE Standard Module Series

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Quectel Wireless Solutions Co., Ltd.

Building 5, Shanghai Business Park Phase III (Area B), No.1016 Tianlin Road, Minhang District, Shanghai 200233, China

Tel: +86 21 5108 6236

Email: info@quectel.com

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About the Document

Revision History

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-	2022-01-06	Braden HE	Creation of the document
1.0	2022-02-17	Braden HE	First official release
1.1	2023-08-31	Joe TU	1. Added the applicable modules EG912U-GL and EG915U series. 2. Added members to <code>ql_datacall_errcode_e</code> enumeration (Chapter 3.3.1.1). 3. Added <code>ql_datacall_set_default_pdn_cfg()</code> (Chapter 3.3.13). 4. Added <code>ql_datacall_get_default_pdn_cfg()</code> (Chapter 3.3.14). 5. Added <code>ql_datacall_get_default_pdn_info</code> (Chapter 3.3.15).

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1 Introduction

Quectel LTE Standard EC200U series and EG91xU family modules support QuecOpen® solution. QuecOpen® is an embedded development platform based on RTOS, which is intended to simplify the design and development of IoT applications. For more information on QuecOpen®, see **document [1]**.

This document explains data call, including API calling process and provides API description on Quectel EC200U series and EG91xU family modules in QuecOpen® solution.

1.1. Applicable Modules

Table 1: Applicable Modules

Module Family	Module
-	EC200U Series
EG91xU	EG912U-GL
	EG915U Series

2 API Calling Process

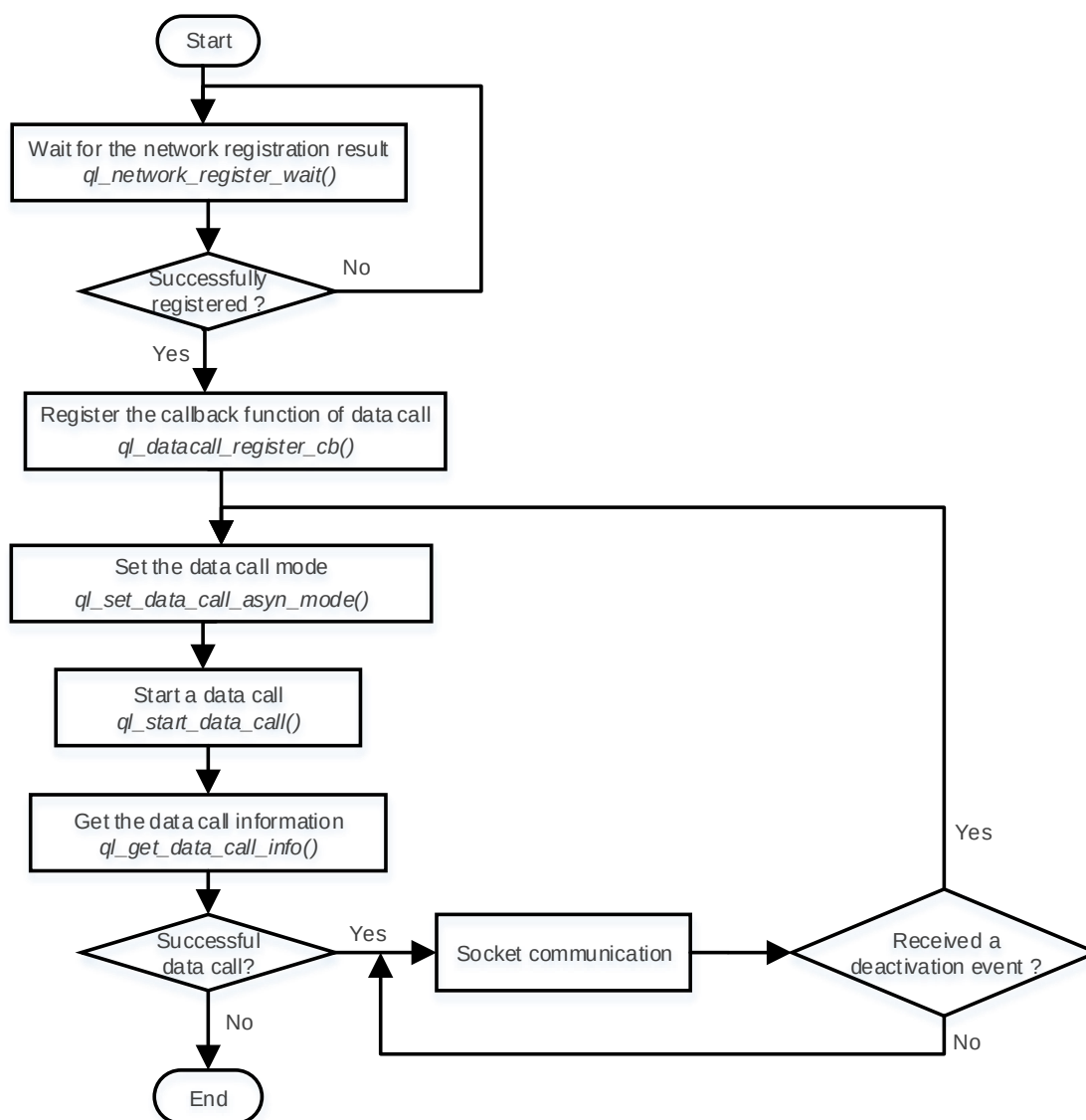


Figure 1: Data Call Process

Data call process includes three parts: network registration, data call and socket establishment.

Network registration is performed automatically and requires no code intervention. You only need to call *ql_network_register_wait()* to wait for the registration to complete. If the current module does not register network, the thread currently calling this function will be blocked until the network registration is successful

or times out. You can also call `ql_nw_register_cb()` to register the callback function for network event to receive `QUEC_NW_DATA_REG_STATUS_IND` event. See **document [2]** for detailed information about `ql_nw_register_cb()`.

PDP context activation can be carried out once the module has successfully registered the network. Current data call supports both synchronous and asynchronous modes, which can be set with `ql_set_data_call_asyn_mode()`. See **Chapter 3.3.2** for detailed information.

Socket communication can be carried out after a successful data call. To help you understand the data call process, `datacall_demo.c`, the example program, is available in `components\ql-application\nw\` directory of QuecOpen SDK.

To start the example program automatically, uncomment `ql_datacall_app_init()` under `ql_init_demo_thread()` in `components\ql-application\init\ql_init.c` file, then compile the firmware version and download it to the module. When the module boots, the example program starts and automatically performs a data call.

```

04:
05: static void ql_init_demo_thread(void *param)
06: {
07:     QL_INIT_LOG("init demo thread enter, param 0x%x", param);
08:
09:     ql_gpio_app_init();
10:     ql_gpioint_app_init();
11:     ql_ledcfg_app_init();
12:
13:     #ifdef QL_APP_FEATURE_AUDIO
14:         ql_audio_app_init();
15:     #endif
16:     #ifdef QL_APP_FEATURE_LCD
17:         ql_lcd_app_init();
18:     #endif
19:     ql_nw_app_init();
20:     //ql_datacall_app_init();
21:     ql_osi_demo_init();
22:
23:     #ifdef QL_APP_FEATURE_FILE
24:         ql_fs_demo_init();
25:     #endif
26:

```

Figure 2: Auto-Start Data Call Example Program

3 Data Call API

This chapter introduces data call API, enumerations and structures.

3.1. Header File

ql_api_datacall.h, the header file of data call API, is in *components\ql-kernel\inc* directory. Unless otherwise specified, the header files mentioned in this document are all in this directory.

3.2. API Overview

Table 2: API Overview

Function	Description
<i>ql_network_register_wait()</i>	Waits for the network registration result.
<i>ql_set_data_call_asyn_mode()</i>	Sets the execution mode of <i>ql_start_data_call()</i> and <i>ql_stop_data_call()</i> .
<i>ql_start_data_call()</i>	Starts a data call.
<i>ql_stop_data_call()</i>	Stops a data call.
<i>ql_get_data_call_info()</i>	Gets data call information.
<i>ql_datacall_register_cb()</i>	Registers the callback function of data call.
<i>ql_datacall_unregister_cb()</i>	Cancels the callback function registered with <i>ql_datacall_register_cb()</i> .
<i>ql_datacall_get_sim_profile_is_active()</i>	Gets the current status of PDP context activation.
<i>ql_bind_sim_and_profile()</i>	Binds the (U)SIM card in use and PDP context ID, and gets <i>sim_cid</i> , the combination of (U)SIM card and PDP context ID.
<i>ql_get_sim_and_profile()</i>	Gets (U)SIM card and PDP context ID from <i>sim_cid</i> .

<code>ql_datacall_set_nat()</code>	Enables NAT.
<code>ql_datacall_get_nat()</code>	Gets the NAT-enabled (U)SIM card and PDP context ID.
<code>ql_datacall_set_default_pdn_cfg()</code>	Sets the default bearer configuration.
<code>ql_datacall_get_default_pdn_cfg()</code>	Gets the default bearer configuration.
<code>ql_datacall_get_default_pdn_info()</code>	Gets the default bearer information.

3.3. API Description

3.3.1. ql_network_register_wait

This function waits for the network registration result.

● Prototype

```
ql_datacall_errcode_e ql_network_register_wait(uint8_t nSim, unsigned int timeout_s)
```

● Parameter

nSim:

[In] (U)SIM card in use. If the module only supports one (U)SIM interface, this parameter must be set to 0.

0 (U)SIM card 1

1 (U)SIM card 2

timeout_s:

[In] Waiting for network registration timeout. Unit: second.

● Return Value

See **Chapter 3.3.1.1** for detailed information.

3.3.1.1. ql_datacall_errcode_e

The data call result codes indicate whether the function is executed successfully. The enumeration of result codes is defined as below:

```
typedef enum
{
    QL_DATACALL_SUCCESS                = 0,
    QL_DATACALL_EXECUTE_ERR            = 1 | QL_DATACALL_ERRCODE_BASE,
    QL_DATACALL_MEM_ADDR_NULL_ERR,
```

```

QL_DATACALL_INVALID_PARAM_ERR,
QL_DATACALL_NW_REGISTER_TIMEOUT_ERR,
QL_DATACALL_CFW_ACT_STATE_GET_ERR    = 5 | QL_DATACALL_ERRCODE_BASE,
QL_DATACALL_REPEAT_ACTIVE_ERR,
QL_DATACALL_REPEAT_DEACTIVE_ERR,
QL_DATACALL_CFW_PDP_CTX_SET_ERR,
QL_DATACALL_CFW_PDP_CTX_GET_ERR,
QL_DATACALL_CS_CALL_ERR                = 10 | QL_DATACALL_ERRCODE_BASE,
QL_DATACALL_CFW_CFUN_GET_ERR,
QL_DATACALL_CFUN_DISABLE_ERR,
QL_DATACALL_NW_STATUS_GET_ERR,
QL_DATACALL_NOT_REGISTERED_ERR,
QL_DATACALL_NO_MEM_ERR                  = 15 | QL_DATACALL_ERRCODE_BASE,
QL_DATACALL_CFW_ATTACH_STATUS_GET_ERR,
QL_DATACALL_SEMAPHORE_CREATE_ERR,
QL_DATACALL_SEMAPHORE_TIMEOUT_ERR,
QL_DATACALL_CFW_ATTACH_REQUEST_ERR,
QL_DATACALL_CFW_ACTIVE_REQUEST_ERR = 20 | QL_DATACALL_ERRCODE_BASE,
QL_DATACALL_ACTIVE_FAIL_ERR,
QL_DATACALL_CFW_DEACTIVE_REQUEST_ERR,
QL_DATACALL_NO_DFTPDN_CFG_CONTEXT,
QL_DATACALL_NO_DFTPDN_INFO_CONTEXT,
}ql_datacall_errcode_e;

```

● Member

Member	Description
<code>QL_DATACALL_SUCCESS</code>	Successful execution.
<code>QL_DATACALL_EXECUTE_ERR</code>	Failed execution.
<code>QL_DATACALL_MEM_ADDR_NULL_ERR</code>	Parameter address is NULL.
<code>QL_DATACALL_INVALID_PARAM_ERR</code>	Invalid parameter.
<code>QL_DATACALL_NW_REGISTER_TIMEOUT_ERR</code>	Network registration timeout.
<code>QL_DATACALL_CFW_ACT_STATE_GET_ERR</code>	Failed to get the status of PDP context activation.
<code>QL_DATACALL_REPEAT_ACTIVE_ERR</code>	Repeatedly activate PDP context ID.
<code>QL_DATACALL_REPEAT_DEACTIVE_ERR</code>	Repeatedly deactivate PDP context ID.
<code>QL_DATACALL_CFW_PDP_CTX_SET_ERR</code>	PDP context ID setting error.
<code>QL_DATACALL_CFW_PDP_CTX_GET_ERR</code>	PDP context ID getting error.

<code>QL_DATACALL_CS_CALL_ERR</code>	Data service operation failure due to an ongoing call.
<code>QL_DATACALL_CFW_CFUN_GET_ERR</code>	Failed to get functionality level.
<code>QL_DATACALL_CFUN_DISABLE_ERR</code>	Data service operation failure due to non-full-functionality.
<code>QL_DATACALL_NW_STATUS_GET_ERR</code>	Failed to get network registration status.
<code>QL_DATACALL_NOT_REGISTERED_ERR</code>	Network unregistered.
<code>QL_DATACALL_NO_MEM_ERR</code>	Failed to request memory.
<code>QL_DATACALL_CFW_ATTACH_STATUS_GET_ERR</code>	Failed to get network attachment status.
<code>QL_DATACALL_SEMAPHORE_CREATE_ERR</code>	Failed to create semaphore.
<code>QL_DATACALL_SEMAPHORE_TIMEOUT_ERR</code>	Waiting for semaphore timeout.
<code>QL_DATACALL_CFW_ATTACH_REQUEST_ERR</code>	Network attachment is rejected.
<code>QL_DATACALL_CFW_ACTIVE_REQUEST_ERR</code>	PDP context activation is rejected.
<code>QL_DATACALL_ACTIVE_FAIL_ERR</code>	Failed to activate PDP context.
<code>QL_DATACALL_CFW_DEACTIVE_REQUEST_ERR</code>	PDP context deactivation is rejected.
<code>QL_DATACALL_NO_DFTPDN_CFG_CONTEXT</code>	The default bearing context is not configured.
<code>QL_DATACALL_NO_DFTPDN_INFO_CONTEXT</code>	No default bearing context information.

3.3.2. `ql_set_data_call_asyn_mode`

This function sets the execution mode of `ql_start_data_call()` (see **Chapter 3.3.3** for details) and `ql_stop_data_call()` (see **Chapter 3.3.4** for details). There are two execution modes: synchronous and asynchronous.

● Prototype

```
ql_datacall_errcode_e ql_set_data_call_asyn_mode(uint8_t nSim, int profile_idx, bool enable)
```

● Parameter

nSim:

[In] (U)SIM card in use. If the module only supports one (U)SIM interface, this parameter must be set to 0.

0 (U)SIM card 1

1 (U)SIM card 2

profile_idx:

[In] PDP context ID. Range: 1–7.

enable:

[In] Execution mode.

- 0 Synchronous mode
- 1 Asynchronous mode

● Return Value

See **Chapter 3.3.1.1** for details.

3.3.3. ql_start_data_call

This function starts a data call. Default execution mode: synchronous mode. To set the execution mode to asynchronous mode, call *ql_set_data_call_asyn_mode()* (see **Chapter 3.3.2** for details). The differences between synchronous mode and asynchronous mode are as follows:

Synchronous mode: The data call result code is returned after the function is executed. *QL_DATACALL_SUCCESS* being returned indicates that the data call is successful, IP address is obtained, and socket communication can be carried out.

Asynchronous mode: The function execution result code is returned after the function is executed. *QL_DATACALL_SUCCESS* being returned does not indicate a successful data call, but only a successful function execution. The registered callback function will inform the upper layer whether the data call is successful via *QUEC_DATACALL_ACT_RSP_IND* event. See **Chapter 3.3.6.1** for detailed information on the callback function and event.

● Prototype

```
ql_datacall_errcode_e ql_start_data_call
(
    uint8_t nSim,
    int profile_idx,
    int ip_version,
    char *apn_name,
    char *username,
    char *password,
    int auth_type
)
```

● Parameter

nSim:

[In] (U)SIM card in use. If the module only supports one (U)SIM interface, this parameter must be set to 0.

- 0 (U)SIM card 1
- 1 (U)SIM card 2

profile_idx:

[In] PDP context ID. Range: 1–7.

ip_version:

[In] IP version.

- 1 IPv4
- 2 IPv6
- 3 IPv4v6

apn_name:

[In] Access point name.

username:

[In] Username.

password:

[In] Password.

auth_type:

[In] Authentication type. See **Chapter 3.3.3.1** for details.

● Return Value

See **Chapter 3.3.1.1** for details.

NOTE

If VoLTE function is enabled, only the PDP context 1–5 can be available during a data call.

3.3.3.1. ql_pdp_auth_type_e

The enumeration of authentication type is defined as below:

```
typedef enum
{
    QL_PDP_AUTH_TYPE_NONE = 0,
    QL_PDP_AUTH_TYPE_PAP,
```

```
QL_PDP_AUTH_TYPE_CHAP,
QL_PDP_AUTH_TYPE_PAP_OR_CHAP,
}ql_pdp_auth_type_e;
```

● Member

Member	Description
<code>QL_PDP_AUTH_TYPE_NONE</code>	No authentication protocols.
<code>QL_PDP_AUTH_TYPE_PAP</code>	PAP.
<code>QL_PDP_AUTH_TYPE_CHAP</code>	CHAP.

NOTE

EC200U series and EG91xU family modules do not support PAP or CHAP automatic authentication. If you need automatic authentication, refer to *nw_demo.c* and activate PDP again based on the reject cause returned by QUEEC_NW_CAUSE_IND event in the callback.

3.3.4. ql_stop_data_call

This function stops a data call. The default execution mode is synchronous mode and you can set the execution mode to asynchronous mode by calling *ql_set_data_call_asyn_mode()* (see **Chapter 3.3.2** for details).

● Prototype

```
ql_datacall_errcode_e ql_stop_data_call(uint8_t nSim, int profile_idx)
```

● Parameter

nSim:

[In] (U)SIM card in use. If the module only supports one (U)SIM interface, this parameter must be set to 0.

- 0 (U)SIM card 1
- 1 (U)SIM card 2

profile_idx:

[In] PDP context ID. Range: 1–7.

● Return Value

See **Chapter 3.3.1.1** for details.

3.3.5. ql_get_data_call_info

This function gets data call information.

● Prototype

```
ql_datacall_errcode_e ql_get_data_call_info(uint8_t nSim, int profile_idx, ql_data_call_info_s *info)
```

● Parameter

nSim:

[In] (U)SIM card in use. If the module only supports one (U)SIM interface, this parameter must be set to 0.

0 (U)SIM card 1

1 (U)SIM card 2

profile_idx:

[In] PDP context ID. Range: 1–7.

info:

[Out] Data call information. See **Chapter 3.3.5.1** for details.

● Return Value

See **Chapter 3.3.1.1** for details.

3.3.5.1. ql_data_call_info_s

The structure of data call information is defined as below:

```
typedef struct
{
    int profile_idx;
    int ip_version;
    struct v4_info v4;
    struct v6_info v6;
}ql_data_call_info_s;
```

● Parameter

Type	Parameter	Description
int	<i>profile_idx</i>	PDP context ID. Range: 1–7.
int	<i>ip_version</i>	IP version. 1 IPv4 2 IPv6

3 IPv4v6

<i>struct v4_info</i>	<i>v4</i>	IPv4 information. See Chapter 3.3.5.2 for details.
<i>struct v6_info</i>	<i>v6</i>	IPv6 information. See Chapter 3.3.5.4 for details.

3.3.5.2. struct v4_info

The structure of IPv4 information is defined as below:

```
struct v4_info
{
    int state;
    struct v4_address_status addr;
};
```

● Parameter

Type	Parameter	Description
int	<i>state</i>	Data call status. 0 No data call 1 Successful data call
<i>struct v4_address_status</i>	<i>addr</i>	IPv4 address information. See Chapter 3.3.5.3 for details.

3.3.5.3. struct v4_address_status

The structure of IPv4 address information is defined as below:

```
struct v4_address_status
{
    ip4_addr_t ip;
    ip4_addr_t pri_dns;
    ip4_addr_t sec_dns;
};
```

● Parameter

Type	Parameter	Description
<i>ip4_addr_t</i>	<i>ip</i>	IPv4 address obtained. See Chapter 3.3.5.3.1 for details.
<i>ip4_addr_t</i>	<i>pri_dns</i>	Primary DNS IPv4 address. See Chapter 3.3.5.3.1 for details.
<i>ip4_addr_t</i>	<i>sec_dns</i>	Secondary DNS IPv4 address. See Chapter 3.3.5.3.1 for details.

3.3.5.3.1. ip4_addr_t

The structure of IPv4 address is defined as below:

```
struct ip4_addr_t
{
    uint32_t addr;
};
```

● Parameter

Type	Parameter	Description
uint32_t	<i>addr</i>	IPv4 address.

3.3.5.4. struct v6_info

The structure of IPv6 information is defined as below:

```
struct v6_info
{
    int state;
    struct v6_address_status addr;
};
```

● Parameter

Type	Parameter	Description
int	<i>state</i>	Data call status. 0 No data call 1 Successful data call
<i>struct v6_address_status</i>	<i>addr</i>	IPv6 address information. See Chapter 3.3.5.5 for details.

3.3.5.5. struct v6_address_status

The structure of IPv6 address information is defined as below:

```
struct v6_address_status
{
    ip6_addr_t ip;
    ip6_addr_t pri_dns;
    ip6_addr_t sec_dns;
};
```

- **Parameter**

Type	Parameter	Description
<i>ip6_addr_t</i>	<i>ip</i>	IPv6 address obtained. See Chapter 3.3.5.5.1 for details.
<i>ip6_addr_t</i>	<i>pri_dns</i>	Primary DNS IPv6 address. See Chapter 3.3.5.5.1 for details.
<i>ip6_addr_t</i>	<i>sec_dns</i>	Secondary DNS IPv6 address. See Chapter 3.3.5.5.1 for details.

3.3.5.5.1. ip6_addr_t

The structure of IPv6 address is defined as below:

```
struct ip6_addr_t
{
    uint32_t addr[4];
};
```

- **Parameter**

Type	Parameter	Description
<i>uint32_t</i>	<i>addr[4]</i>	IPv6 address.

3.3.6. ql_datacall_register_cb

This function registers the callback function of data call. Whether in asynchronous or synchronous mode, the module needs to register callback function of data call to report the event of deactivation or de-attachment from the network, and the module can initiate the data call process again after receiving this event.

- **Prototype**

```
ql_datacall_errcode_e ql_datacall_register_cb(uint8_t nSim, int profile_idx, ql_datacall_callback datacall_cb, void *ctx)
```

- **Parameter**

nSim:

[In] (U)SIM card in use. If the module only supports one (U)SIM interface, this parameter must be set to 0.

- 0 (U)SIM card 1
- 1 (U)SIM card 2

profile_idx:

[In] PDP context ID. Range: 1–7.

datacall_cb:

[In] Callback function to be registered. See **Chapter 3.3.6.1** for details.

ctx:

[In] Pointer to the passed parameters of the callback function.

● Return Value

See **Chapter 3.3.1.1** for details.

3.3.6.1. ql_datacall_callback

This callback function reports data call event.

● Prototype

```
typedef void (*ql_datacall_callback)(uint8_t nSim, unsigned int ind_type, int profile_idx, bool result, void *ctx)
```

● Parameter

nSim:

[In] (U)SIM card in use. If the module only supports one (U)SIM interface, this parameter must be set to 0.

- 0 (U)SIM card 1
- 1 (U)SIM card 2

ind_type:

[In] Data call event type.

<code>QUEC_DATACALL_ACT_RSP_IND</code>	Response event of PDP activation result in asynchronous mode
<code>QUEC_DATACALL_DEACT_RSP_IND</code>	Response event of PDP deactivation result in asynchronous mode
<code>QUEC_DATACALL_PDP_DEACTIVE_IND</code>	Event of PDP deactivation or de-attachment by the network

profile_idx:

[In] PDP context ID. Range: 1–7.

result:

[In] Result of PDP context activation and deactivation.

- 0 Failed
- 1 Successful

ctx:

[In] Pointer to the passed parameters of the callback function.

- **Return Value**

None

3.3.7. ql_datacall_unregister_cb

This function cancels the callback function registered with *ql_datacall_register_cb()*. The callback function no longer receives data call events once it is cancelled.

- **Prototype**

```
ql_datacall_errcode_e ql_datacall_unregister_cb(uint8_t nSim, int profile_idx, ql_datacall_callback
datacall_cb, void *ctx)
```

- **Parameter**

nSim:

[In] (U)SIM card in use. If the module only supports one (U)SIM interface, this parameter must be set to 0.

0 (U)SIM card 1

1 (U)SIM card 2

profile_idx:

[In] PDP context ID. Range: 1–7.

datacall_cb:

[In] Callback function to be cancelled. See **Chapter 3.3.6.1** for details.

ctx:

[In] Pointer to the passed parameters of the callback function.

- **Return Value**

See **Chapter 3.3.1.1** for details.

3.3.8. ql_datacall_get_sim_profile_is_active

This function gets the current status of PDP context activation.

- **Prototype**

```
bool ql_datacall_get_sim_profile_is_active(uint8_t nSim, int profile_idx)
```

● Parameter

nSim:

[In] (U)SIM card in use. If the module only supports one (U)SIM interface, this parameter must be set to 0.

- 0 (U)SIM card 1
- 1 (U)SIM card 2

profile_idx:

[In] PDP context ID. Range: 1–7.

● Return Value

- 0 Not activated
- 1 Activated

3.3.9. ql_bind_sim_and_profile

This function binds the (U)SIM card in use and PDP context ID, and gets *sim_cid*, the combination of (U)SIM card and PDP context ID.

● Prototype

```
ql_datacall_errcode_e ql_bind_sim_and_profile(uint8_t nSim, int profile_idx, uint16_t *sim_cid)
```

● Parameter

nSim:

[In] (U)SIM card in use. If the module only supports one (U)SIM interface, this parameter must be set to 0.

- 0 (U)SIM card 1
- 1 (U)SIM card 2

profile_idx:

[In] PDP context ID. Range: 1–7.

sim_cid:

[Out] Combination of (U)SIM card and PDP context ID. Calculation formula: $nSim \ll 5 \mid profile_idx$

● Return Value

See **Chapter 3.3.1.1** for details.

3.3.10. ql_get_sim_and_profile

This function gets (U)SIM card and PDP context ID from *sim_cid*.

- **Prototype**

```
ql_datacall_errcode_e ql_get_sim_and_profile(uint16_t sim_cid, uint8_t *nSim, int *profile_idx)
```

- **Parameter**

sim_cid:

[In] Combination of (U)SIM card and PDP context ID.

nSim:

[Out] (U)SIM card in use. If the module only supports one (U)SIM interface, this parameter must be set to 0.

0 (U)SIM card 1

1 (U)SIM card 2

profile_idx:

[Out] PDP context ID. Range: 1–7.

- **Return Value**

See **Chapter 3.3.1.1** for details.

3.3.11. ql_datacall_set_nat

This function enables NAT. The configuration takes effect after the module is rebooted.

- **Prototype**

```
ql_datacall_errcode_e ql_datacall_set_nat(uint32_t sim_profile_list)
```

- **Parameter**

sim_profile_list:

[In] List of (U)SIM card and PDP context ID combinations. The lower 16 bits belong to (U)SIM card 1 and the higher 16 bits belong to (U)SIM card 2. Each bit corresponds to a PDP context ID.

- **Return Value**

See **Chapter 3.3.1.1** for details.

3.3.12. ql_datacall_get_nat

This function gets the NAT-enabled (U)SIM card and PDP context ID.

- **Prototype**

```
ql_datacall_errcode_e ql_datacall_get_nat(uint32_t *sim_profile_list)
```

- **Parameter**

sim_profile_list:

[Out] List of (U)SIM card and PDP context ID combinations. The lower 16 bits belong to (U)SIM card 1 and the higher 16 bits belong to (U)SIM card 2. Each bit corresponds to a PDP context ID.

- **Return Value**

See **Chapter 3.3.1.1** for details.

3.3.13. ql_datacall_set_default_pdn_cfg

This function sets the default bearer configuration.

- **Prototype**

```
ql_datacall_errcode_e ql_datacall_set_default_pdn_cfg(uint8_t nSim, ql_data_call_default_pdn_cfg_s* ctx)
```

- **Parameter**

nSim:

[In] (U)SIM card in use. If the module only supports one (U)SIM interface, this parameter must be set to 0.

0 (U)SIM card 1

1 (U)SIM card 2

ctx:

[In] Configuration information. See **Chapter 3.3.13.1** for details.

- **Return Value**

See **Chapter 3.3.1.1** for details.

3.3.13.1. ql_data_call_default_pdn_cfg_s

The structure of the default bearer configuration is defined as below:

```
typedef struct
{
    int ip_version;
    char apn_name[APN_LEN_MAX];
    char username[USERNAME_LEN_MAX];
    char password[PASSWORD_LEN_MAX];
    int auth_type;
}ql_data_call_default_pdn_cfg_s
```

● Parameter

Type	Parameter	Description
<i>int</i>	<i>ip_version</i>	IP type. 1 IPv4 2 IPv6 3 IPv4v6
<i>char</i>	<i>apn_name</i>	APN name.
<i>char</i>	<i>username</i>	User name.
<i>char</i>	<i>password</i>	Password.
<i>Int</i>	<i>auth_type</i>	Authentication type. See Chapter 3.3.3.1 for details

3.3.14. ql_datacall_get_default_pdn_cfg

The function gets the default bearer configuration.

● Prototype

```
ql_datacall_errcode_e ql_datacall_get_default_pdn_cfg(uint8_t nSim, ql_data_call_default_pdn_cfg_s* ctx)
```

● Parameter

nSim:

[In] (U)SIM card in use. If the module only supports one (U)SIM interface, this parameter must be set to 0.

- 0 (U)SIM card 1
- 1 (U)SIM card 2

ctx:

[Out] Configuration information. See **Chapter 3.3.13.1** for details.

- **Return Value**

See **Chapter 3.3.1.1** for details.

3.3.15. ql_datacall_get_default_pdn_info

The function gets the default bearer information.

- **Prototype**

```
ql_datacall_errcode_e ql_datacall_get_default_pdn_info(uint8_t nSim,
ql_data_call_default_pdn_info_s* ctx)
```

- **Parameter**

nSim:

[In] (U)SIM card in use. If the module only supports one (U)SIM interface, this parameter must be set to 0.

0 (U)SIM card 1

1 (U)SIM card 2

ctx:

[Out] The bearer information. See **Chapter 3.3.15.1** for details.

- **Return Value**

See **Chapter 3.3.1.1** for details.

3.3.15.1. ql_data_call_default_pdn_info_s

The structure of the default bearer information is defined as below:

```
typedef struct
{
uint8_t          ip_version;
ip4_addr_t      ipv4;
ip6_addr_t      ipv6;
char            apn_name[APN_LEN_MAX];
}ql_data_call_default_pdn_info_s
```

- Parameter

Type	Parameter	Description
<i>uint8_t</i>	<i>ip_version</i>	IP type. 1 IPv4 2 IPv6 3 IPv4v6
<i>ip4_addr_t</i>	<i>ipv4</i>	IPv4 information. See Chapter 3.3.5.2 for details.
<i>ip6_addr_t</i>	<i>ipv6</i>	IPv6 information. See Chapter 3.3.5.4 for details.
<i>char</i>	<i>apn_name</i>	APN name.

4 Appendix References

Table 3: Related Documents

Document Name
[1] Quectel_EC200U&EG91xU_Series_QuecOpen_CSDK_Quick_Start_Guide
[2] Quectel_EC200U&EG91xU_Series_QuecOpen_Cellular_Network_Information_API_Reference_Manual

Table 4: Terms and Abbreviations

Abbreviation	Description
API	Application Programming Interface
CHAP	Challenge Handshake Authentication Protocol
DNS	Domain Name Server
ID	Identifier
IPv4	Internet Protocol Version 4
IPv6	Internet Protocol Version 6
NAT	Network Address Translation
PAP	Password Authentication Protocol
PDP	Packet Data Protocol
SDK	Software Development Kit
(U)SIM	(Universal) Subscriber Identity Module